

$$\begin{aligned}v(t) &= 170 \sin(120\pi t) \\v'(t) &= 170 \cos(120\pi t)(120\pi) \\&= 20400 \cos(120\pi t)\end{aligned}$$

Critical numbers:

$$0 = \cos(120\pi t)$$

Case 1:

$$\begin{aligned}120\pi t &= \arccos 0 \\120\pi t &= \frac{\pi}{2} \\t &= \frac{\pi}{2 * 120\pi} \\t &= \frac{1}{240}\end{aligned}$$

Case 2:

$$\begin{aligned}120\pi t &= 2\pi - \arccos 0 \\120\pi t &= 2\pi - \frac{\pi}{2} \\t &= \frac{3\pi}{2 * 120\pi} \\t &= \frac{1}{80}\end{aligned}$$